

Integration of Federated Learning IDS and FHE in IoV Environments

1. Introduction

The FedIoV framework presents a secure, scalable, and adaptive Intrusion Detection System (IDS) tailored for real-time Internet of Vehicles (IoV) environments [cite: Fed-lov.pdf]. Conventional centralized IDS architectures fail to meet the strict latency, privacy, and scalability demands of decentralized vehicular networks, exposing systems to adversarial interventions [cite: Fed-lov.pdf]. FedIoV leverages a lightweight *KANConvNet* classifier, utilizing the Kolmogorov-Arnold representation theorem to process complex vehicular dynamics efficiently [cite: Fed-lov.pdf]. To guarantee robustness against non-IID data distributions and adversarial/Byzantine updates, the architecture utilizes a two-stage aggregation pipeline consisting of TOPSIS-based dimensionality reduction followed by Multi-Krum robust clustering [cite: Fed-lov.pdf].

Integrating Fully Homomorphic Encryption (FHE) with the FedIoV architecture is crucial for establishing zero-trust collaborative environments. While Federated Learning (FL) intrinsically prevents the sharing of raw sensor telemetry by exchanging only model parameters [cite: Fed-lov.pdf], the gradients themselves remain vulnerable to model inversion and membership inference attacks at the central aggregator. FHE enables the central server to aggregate model updates while they remain strictly encrypted, ensuring true end-to-end mathematical privacy without exposing local fleet operational patterns.

FedIoV's empirical validation employs five core benchmark datasets: Car-Hacking (CAN bus anomalies), VeReMi (VANET misbehavior), CICIDS-2017 (multi-layer network threats), IEEE VTC-CAN (embedded spoofing), and CarNet (multi-sensor context) [cite: Fed-lov.pdf].

2. Detailed Dataset Analysis

2.1. Car-Hacking Dataset

Extracted from Hyundai vehicles, this dataset encompasses Controller Area Network (CAN) bus traffic representing standard operations alongside DoS, fuzzy, spoofing, and replay attacks [cite: Fed-lov.pdf]. It prioritizes timing, message authenticity, and signal integrity [cite: Fed-lov.pdf].

Dataset	Total Features	Important Features	Removable Features	Why Important	Why Removable
Car-Hacking	12 (Approx)	CAN ID, DLC, Payload (DATA_0 to DATA_7)	Absolute Timestamp	Core identifiers for physical ECU states and injection anomalies.	Highly variant; induces severe model overfitting on exact time values rather than sequential patterns.

2.2. VeReMi Dataset

The VeReMi dataset simulates misbehavior in Vehicle-to-Vehicle (V2V) communications (VANETs), concentrating heavily on position falsification and deceptive messaging [cite: Fed-lov.pdf].

Dataset	Total Features	Important Features	Removable Features	Why Important	Why Removable
VeReMi	~15	Position, Speed, Heading, Acceleration	Raw Node IDs, RSSI (if unstable)	Critical for detecting kinematic impossibility and falsified trajectories.	Node IDs prevent generalization across diverse fleets; raw RSSI is environmentally biased.

2.3. CICIDS-2017 Dataset

Encompassing multi-layer networking attacks (DDoS, brute-force, infiltration), CICIDS-2017 evaluates general V2I and V2N communication layer threats [cite: Fed-lov.pdf].

Dataset	Total Features	Important Features	Removable Features	Why Important	Why Removable
CICIDS-2017	80+	Flow Bytes/s, Packet Length Mean, TCP Flags	Source/Dest IP, Source/Dest Port	Captures network congestion, volumetric anomalies, and session hijacking.	IP/Port specific rules fail completely against zero-day endpoints and dynamically assigned IoV nodes.

2.4. IEEE VTC-CAN & CarNet Datasets

IEEE VTC-CAN focuses on DoS and spoofing inside the CAN protocol for embedded systems with tight latency constraints [cite: Fed-lov.pdf]. CarNet provides multi-sensor telemetry (GPS, LiDAR, kinematics) typical of autonomous V2X situations [cite: Fed-lov.pdf].

Dataset	Total Features	Important Features	Removable Features	Why Important	Why Removable
VTC-CAN	~12	CAN ID, Payload Vectors	Timestamps	Low-level anomaly detection in embedded hardware.	Prevents temporal overfitting.
CarNet	30+	Fused coordinates, Relative velocity	Absolute GPS coords	Essential for contextual cross-sensor anomaly detection.	Absolute coords limit geographic generalization.

2.5. Integration of CICIoV2024 Dataset

Can CICIoV2024 be added? Yes. CICIoV2024 is highly compatible with the unified architecture, focusing heavily on intra-vehicular communications (CAN bus) extracted from a 2019 Ford vehicle. It provides modernized DoS and distinct spoofing vectors (GAS, RPM, SPEED, STEERING_WHEEL).

Would it improve generalization? Incorporating CICIoV2024 significantly boosts model robustness against newer vehicle architectures and specific mechanical spoofing vectors. However, the raw dataset suffers from an "accuracy trap" due to massive repetitive CAN frame redundancy. Strict de-duplication must be enforced so the model learns true zero-day attack signatures rather than over-represented benign duplicates.

Modifications needed: The dataset requires schema alignment. Features such as [DATA_0](#) through [DATA_7](#) and the Arbitration ID must be normalized and embedded similarly to the Car-Hacking integration pipeline. Strict deduplication and ANOVA-based feature selection should be prioritized over full-dimensional deep learning to prevent starvation in minority attack classes.

3. Feature Engineering and Optimization

Optimizing the feature space is paramount for balancing IDS performance with the severe computational limitations of FHE arithmetic.

- **Important Features for IDS:** Payload data bytes, protocol IDs, relative kinematics (acceleration, heading differentials), and volumetric flow statistics.
- **Features to Retain for Unified Training:** Normalized continuous channels (Speed, Yaw rate) and discrete channels (CAN IDs/V2X beacons) resampled to a common 10 Hz grid using zero-order hold and linear interpolation [cite: Fed-lov.pdf].
- **Features to Discard (Pre-FL):** Absolute timestamps, raw GPS coordinates, IPs, and MAC addresses. These cause strict domain-overfitting and bloat the FHE polynomial dimensions without adding conceptual value.
- **FHE Suitability:** FHE schemes efficiently process additions and multiplications on bounded real numbers or integers. Features must undergo Z-score standardization using local statistics and continuous features must be clipped at the 0.5th/99.5th percentiles to restrict outlier magnitudes [cite: Fed-lov.pdf]. Categorical variables (CAN IDs) should utilize low-dimensional embeddings rather than expansive one-hot encoding to preserve FHE ciphertext compactness.

4. Unified Feature Space for IDS + FHE Integration

FedLoV's preprocessing divides diverse modalities into windows (length 64, stride 16) and combines them into a fused fixed-size 46-dimensional input vector for the KANConvNet [cite: Fed-lov.pdf]. To accommodate FHE, a constant feature vector size is strictly required because changing parameter dimensions breaks homomorphic evaluation keys and polynomial moduli.

Unified FHE Feature Sub-Space (46 Dimensions)	Allocation / Methodology
Dimensions 0 - 9 (CAN Bus / In-Vehicle)	Normalized Arbitration ID embeddings, aggregated Payload statistics (e.g., mean/ variance of DATA_0 to DATA_7), Frame frequency.
Dimensions 10 - 24 (Kinematics & Telemetry)	Relative speed, Delta-heading, acceleration, yaw rate, relative distance to nearest V2V nodes.
Dimensions 25 - 39 (Network / V2N)	Flow rates, protocol type embeddings, mean packet lengths, TCP flag frequencies.
Dimensions 40 - 45 (Context & Flags)	Binary masks for missing packets (to handle non-overlapping datasets natively), cooperative V2X flags, fused reliability scores.

When datasets lack overlapping features (e.g., Car-Hacking has no V2X kinematics), missing channels are zero-padded, and a binary mask vector is populated to indicate data absence, preventing the model from updating gradients based on empty modalities [cite: Fed-lov.pdf].

5. FHE Integration Analysis

Integrating FHE directly addresses the "honest-but-curious" server threat model [cite: Fed-lov.pdf], but introduces profound cryptographic friction:

- **Ciphertext Expansion & Overhead:** FHE ciphertext expansion is typically 10x–100x the plaintext size. The 75 KB communication footprint per round observed in FedLoV [cite: Fed-lov.pdf] could balloon to several megabytes, challenging cellular vehicular uplinks.
- **Algorithm Suitability:** *KANConvNet* relies on Kolmogorov-inspired non-linear activation functions applied at the network's edges prior to summation [cite: Fed-lov.pdf]. FHE cannot inherently evaluate non-polynomial non-linearities. These activations must be approximated using low-degree Taylor or Chebyshev polynomials. Shallow, tree-based models or linear classifiers are highly FHE-friendly, but sacrifice accuracy.
- **CKKS Scheme Recommendation:** The Cheon-Kim-Kim-Song (CKKS) scheme is optimal as it supports approximate arithmetic on real-number arrays, fitting seamlessly with gradient updates and Z-score normalized inputs.
- **Cost Reduction:** Encrypted inference at the edge is unfeasible for the sub-60 ms latency constraint [cite: Fed-lov.pdf]. Instead, FHE should be strictly reserved for **Secure Aggregation** of the gradients during the FL rounds. Vehicles perform plain-text local inference and training, encrypt only the model deltas, and transmit them to the server.

6. Federated Learning Considerations

The core of FedLoV relies on filtering anomalous updates to guarantee resilience under adversarial conditions (maintaining ~96.3% accuracy under 50% adversarial ratios) [cite: Fed-lov.pdf].

Impact of Feature Reduction: The server computes weighted normalizations using the TOPSIS method to reduce dimensionality before aggregation [cite: Fed-lov.pdf]. Under FHE, performing TOPSIS requires homomorphic division and sorting, which is exceptionally deep and computationally destructive. *Solution:* Clients must compute local TOPSIS scores or perform localized masking prior to encryption.

Multi-Krum Filtering: Multi-Krum evaluates pairwise squared Euclidean distances across client models to reject poisoned updates [cite: Fed-lov.pdf]. Computing $O(N^2)$ distances on homomorphically encrypted vectors scales catastrophically. To integrate FHE, Multi-Krum must be substituted with a Secure Multi-Party Computation (SMPC) protocol or a simplified secure masking sum, otherwise, aggregation delays will breach the 120 ms global aggregation cycle limit [cite: Fed-lov.pdf]. FHE integration ensures adversarial robustness against eavesdropping, but complicates mathematical outlier filtering.

7. Final Recommendations

- **Dataset Strategy:** Fuse Car-Hacking, CICIDS-2017, CarNet, and CICIoV2024. Perform strict de-duplication to prevent the accuracy trap, focusing purely on zero-day pattern recognition.
- **Unified Dimension:** Enforce a strict 46-dimensional representation aligned with *KANConvNet*'s input architecture. Missing modalities must be zero-padded and tracked via binary masks.
- **FHE Architecture:** Deploy the **CKKS** encryption scheme limited strictly to Gradient Aggregation. Real-time inference must remain unencrypted locally to honor the 10 ms per packet batch processing capability and sub-joule energy footprint [cite: Fed-IoV.pdf].
- **Polynomial Adaptations:** Replace complex Kolmogorov-Arnold non-linearities with 2nd or 3rd-degree polynomial approximations if FHE evaluation across the network depth is strictly mandated.
- **Deployment Approach:** Utilize Hierarchical Aggregation. Roadside Units (RSUs) should perform local partial aggregation and basic TOPSIS filtering to compress data before forwarding homomorphically encrypted gradients to the central cloud, preserving low latency and massive IoV scalability.